Anomaly Detection

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Lecture 05.1 (v4.0.0)


Why did you exclude all these responses?
We define outlier as someone who doesn't like our program
freshspectrum.com

Signposting

- **Anomaly detection**: finding the observations that do not fit the pattern of the rest.
- Sometimes the anomalies are **the thing we want**:
  - fraud, intrusions, machine faults, rare diseases, new astrophysical objects.
- Sometimes they are **the thing we want to get rid of**:
  - data errors and contamination (see Workshop 05.3 on missingness).
- Today:
  - What is an anomaly, and what kinds are there?
  - The main families of methods, and how they reuse the models we have already seen,
  - Turning scores into decisions, and evaluating detectors.

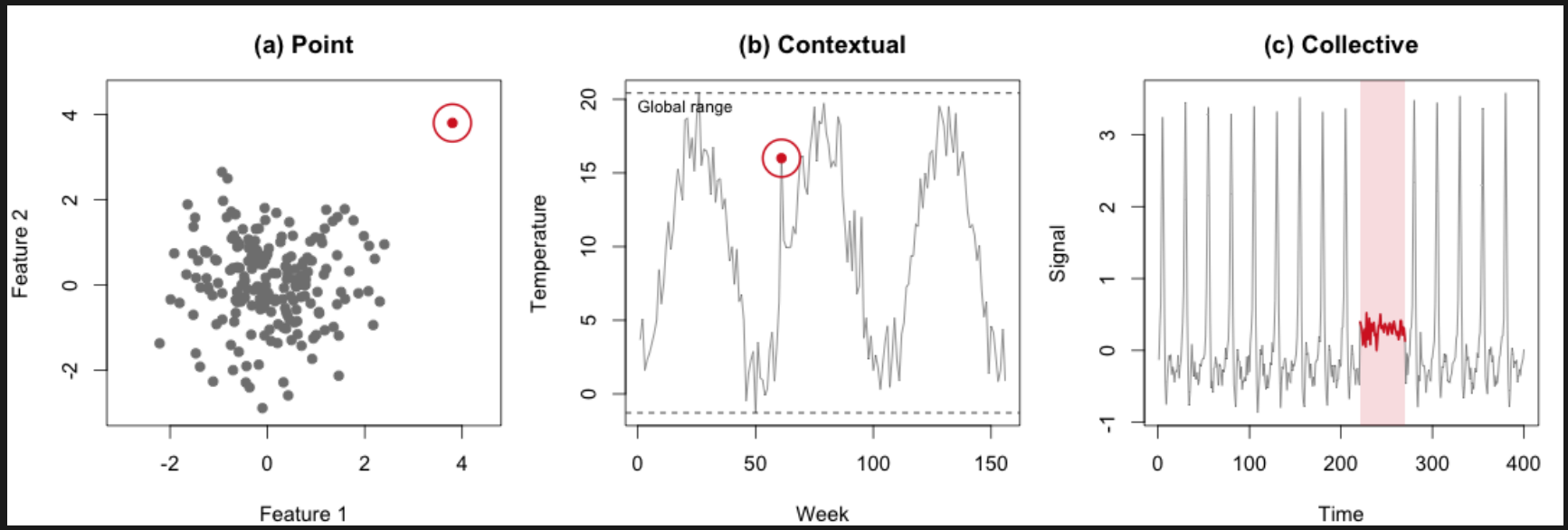
What is an anomaly?

- "An outlier is an observation which deviates so much from the other observations as to arouse suspicions that it was generated by a different mechanism." (Hawkins, 1980)
- The key word is **mechanism**: an anomaly is defined relative to a model of **normal** behaviour.
- Related terms, used loosely and inconsistently in the literature:
  - **Outlier**: often used for anomalies in a static dataset,
  - **Novelty**: a new observation unlike the training data,
  - **Noise**: unusual values that are *not* of interest.
- There is no model-free definition of "unusual". Every method encodes a choice of what normal looks like.

Types of anomaly

1. **Point anomalies**: a single observation is unusual on its own.
   - A £9,000 transaction on a card that usually spends £30.
2. **Contextual (conditional) anomalies**: unusual *given* some context.
   - 30°C is normal in July and anomalous in January.
   - Context variables (time, location) are treated differently from behavioural variables.
3. **Collective anomalies**: a group of observations is unusual together, though each one may look normal.
   - A flat segment in an ECG trace; a coordinated sequence of network requests.

Types of anomaly



Problem settings

- What labels do we have? This decides which methods apply.
- **Supervised**: labelled normal *and* anomalous examples.
  - This is imbalanced classification.
  - Rarely possible: **new kinds** of anomaly keep appearing!
- **Semi-supervised** (novelty detection): training data are normal only, assumed clean.
  - Model normal, and flag departures from it.
- **Unsupervised** (outlier detection): unlabelled data, contaminated with an unknown number of anomalies.
  - Model the bulk, and flag the rest.

The general recipe

1. Build a model of **normal**, $\hat{f}$, from data.
2. Define an **anomaly score** $s(x)$: larger means more unusual.
3. Choose a **threshold** τ, and flag x if $s(x) > \tau$.
4. **Investigate** what was flagged.

- Methods differ mainly in step 1, the notion of normal:
  - **Probabilistic**: normal points have high density,
  - **Proximity**: normal points have close neighbours,
  - **Reconstruction**: normal points lie near a low-dimensional structure,
  - **Isolation**: normal points are hard to separate from the rest,
  - **Boundary**: normal points lie inside a learned region.

The general recipe



Univariate methods

- **z-score**: $z_i = (x_i - \bar{x})/s$, flag $|z_i| > 3$.
  - The mean and standard deviation are **themselves pulled by outliers**.
- **Robust z-score**, medians, quantiles, etc.
- Simple, fast, and interpretable, and they should always be your first look.

Model-based residuals

- Any predictive model gives an anomaly score through its **residuals**.
- **Regression** (Lecture 02.1): large residuals, high **leverage**, and **Cook's distance** measure different things.
  - A high-leverage point can pull the fit towards itself and end up with a small residual.
- **Time series**: forecast $\hat{x}_t$ from the past and score $|x_t - \hat{x}_t|$.
  - This is how **contextual anomalies** are usually handled: the model supplies the context.
- The same caveat as before: fit the model robustly, or the anomalies distort what "normal" means.

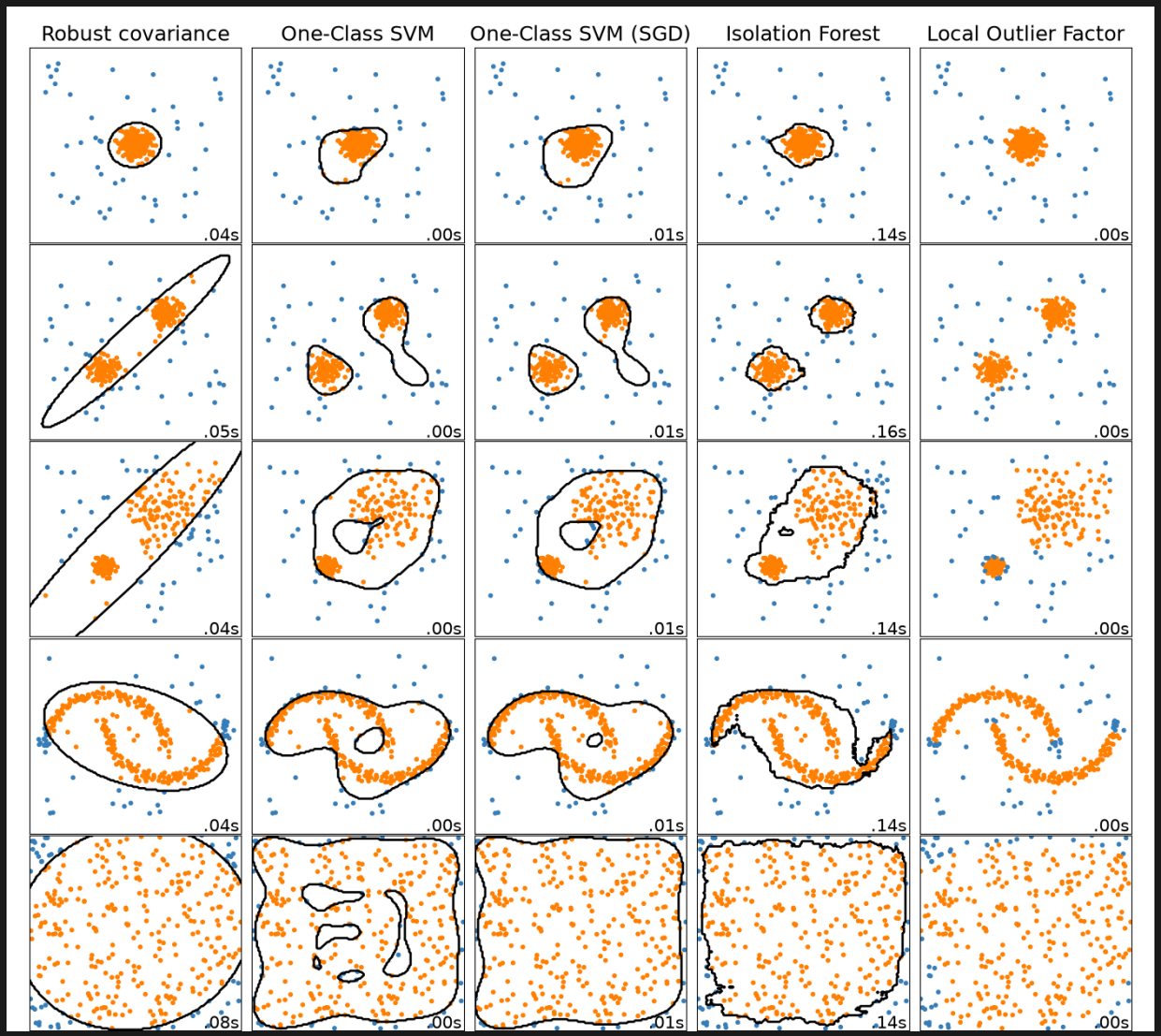
Measuring "Unusual" with models

- Many modelling paradigms **explicitly handle outliers**. Some examples:
- Regression:
  - **leverage** of each point (not always the same as outliers)
  - **Robust regression** methods fit better in the presence of outliers
- **Density-based** clustering (DBSCAN)
  - Points in low density regions may be outliers
  - An empirical p-value can be constructed from the set of points in lower-density regions.
- **Isolation Forests**
  - Random Forest-based technique.
  - Based on identifying "points that are easy to distinguish with a decision tree".
- Many other methods offer $Pr(data|model)$.

No single best method

- Each method's idea of normal suits some data geometries and fails on others.
- Large benchmarks (e.g. ADBench, 2022) find that **no unsupervised method dominates**, and that simple methods are hard to beat.
- Practical advice:
  - Try several methods of different kinds. Agreement between them is informative.
  - Standardise features, since every distance-based method depends on scaling.
  - Use your knowledge of the domain to choose features, which often matters more than the choice of algorithm.

Scikit-Learn Anomaly Detection Examples



From scores to decisions

- Scores are rarely calibrated: an LOF of 1.8 or an Isolation Forest score of 0.6 has no absolute meaning.
- Common ways to set τ:
  - **Contamination**: flag the top α fraction (the `contamination` argument in scikit-learn). This assumes you know α.
  - **Top-k**: flag as many as you can afford to investigate.
  - **Reference distribution**: e.g. the χ^2_p quantile for Mahalanobis distances.
- **Conformal p-values** (semi-supervised): score a held-out calibration set of n normal points, then

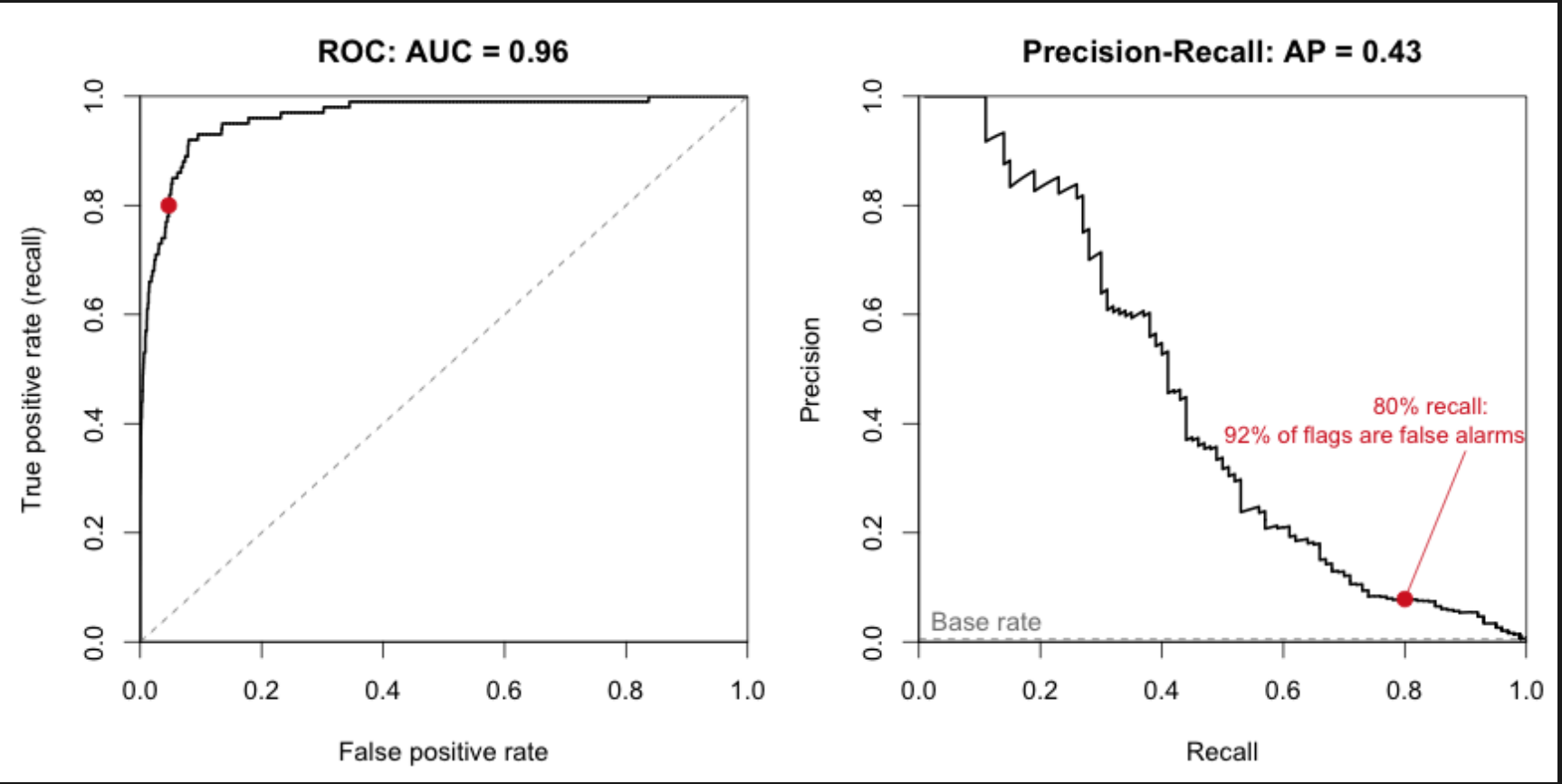
$$p(x) = \frac{1 + \#\{i : s(x_i) \geq s(x)\}}{n + 1}.$$

  - $p(x)$ is a valid p-value whenever x is exchangeable with the calibration data, **whatever** detector produced s.
  - Testing many points means **multiple testing**. Benjamini–Hochberg on conformal p-values controls the false discovery rate (Bates et al., 2023).

Evaluating a detector

- Labels are rare, so evaluation is hard, and often done on:
  - benchmark datasets whose "anomalies" are a rare class,
  - **synthetic anomalies** injected into real data, which may not look like real anomalies.
- Anomaly detection is an extreme case of **class imbalance**:
  - ROC AUC can look excellent while almost everything flagged is a false alarm,
  - **Precision–Recall** curves and **precision@k** are more informative (Lecture 01.2).
- The real test is operational: of the flagged cases a human investigated, how many were worth it?

Evaluating a detector



Anomaly, error, or discovery?

- Detection is only half the job. Every flagged point needs a **decision**:
  - a **data error**: correct it or remove it (Workshop 05.3),
  - a **genuine rare event**: often exactly what you were looking for,
  - a sign that your **model of normal is wrong**: revise the model, not the data.
- Removing anomalies without understanding them can remove the most important signal in the data.
  - The ozone hole was reportedly delayed in discovery because extreme satellite readings were being flagged as bad data.

Reflection

- Why is there no model-free definition of an anomaly? What does each method family assume normal looks like?
- When would you prefer LOF to kNN distance? Isolation Forest to a GMM?
- If anomalies are 1% of your data, what does a detector with 99% accuracy tell you?
- Your training data are contaminated with an unknown fraction of anomalies. Which methods are most and least affected?
- How would you design an evaluation if you had no labelled anomalies at all?

Signposting

- The workshop compares these methods on simulated and real data.
- Further Reading:
  - Chandola, Banerjee & Kumar (2009), "Anomaly detection: A survey", ACM Computing Surveys 41(3). The standard overview.
  - Aggarwal (2017), Outlier Analysis, 2nd ed., Springer. The textbook.
  - Ruff et al. (2021), "A Unifying Review of Deep and Shallow Anomaly Detection", Proceedings of the IEEE.
  - Han et al. (2022), "ADBench: Anomaly Detection Benchmark", NeurIPS.
- Key papers: Breunig et al. (2000) *LOF*; Liu, Ting & Zhou (2008) *Isolation Forest*; Rousseeuw & Van Driessen (1999) *FAST-MCD*; Bates et al. (2023) *Testing for outliers with conformal p-values*, Annals of Statistics.
- Software: scikit-learn (`sklearn.neighbors`, `sklearn.ensemble`, `sklearn.covariance`), PyOD, R packages `robustbase`, `dbscan`, `isotree`.